

MA374 Financial Engineering Lab

Lab 09 Report

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1 Introduction

In this lab, we analyze stock market data from NSE and BSE. The objective is to compute historical volatility using one month of data and use it in option pricing models in later parts.

2 Methodology

The historical volatility is computed as follows:

- Daily returns are calculated using:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

- The standard deviation of daily returns is computed as σ_d .
- Annualized volatility is obtained using:

$$\sigma = \sigma_d \times \sqrt{252}$$

3 Results

3.1 NSE Data

Stock	Volatility
NIFTY ($^N SEI$)	0.0736
RELIANCE	0.1056
TCS	0.1385
INFY	0.1767
HDFCBANK	0.1043

Stock	Volatility
ICICIBANK	0.1054
LT	0.1207
ITC	0.0849
SBIN	0.1471
BHARTIARTL	0.1636
HINDUNILVR	0.2173
IDEA	0.4115
YESBANK	0.1760
SUZLON	0.2257
PNB	0.2844
IRFC	0.4758
FEDERALBNK	0.1406
BHEL	0.2586
SAIL	0.2933
IDFCFIRSTB	0.1987
UCOBANK	0.1800

3.2 BSE Data

Stock	Volatility
SENSEX ($^B SESN$)	0.0695
RELIANCE	0.1051
TCS	0.1388
INFY	0.1742
HDFCBANK	0.1016
ICICIBANK	0.1045
LT	0.1193
ITC	0.0854
SBIN	0.1471
BHARTIARTL	0.1599
HINDUNILVR	0.2132
IDEA	0.4060
YESBANK	0.1795
SUZLON	0.2251
PNB	0.2889
IRFC	0.4703
FEDERALBNK	0.1432
BHEL	0.2610
SAIL	0.2924
IDFCFIRSTB	0.1987
UCOBANK	0.1814

4 Observations

- Large-cap stocks like RELIANCE, HDFCBANK, and ICICIBANK show relatively low volatility.
- Mid and small-cap stocks such as IRFC, IDEA, and SUZLON exhibit significantly higher volatility.
- The indices (NIFTY and SENSEX) have the lowest volatility, indicating diversification effects.
- Banking and PSU stocks tend to have moderate to high volatility.

5 Conclusion

The analysis shows that volatility varies significantly across stocks depending on market capitalization and sector. These volatility estimates will be used in option pricing models in the next parts of the lab.